Mathematics I

Week 5

Graded Assignment Practice Session

Plan for this session

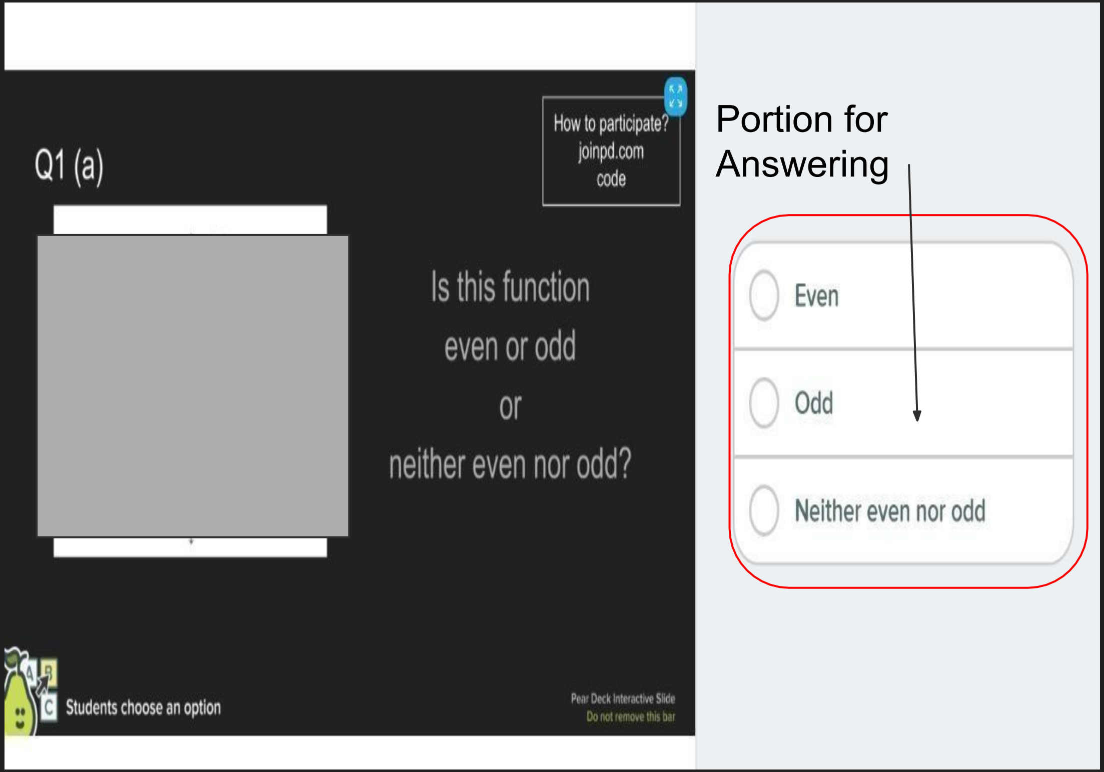
- How to join?
    - Join on webex - click on link sent to you
    - Join on pear deck - joinpd.com (enter code: see top right on webex)
    - Keep a notebook and pen ready for solving problems

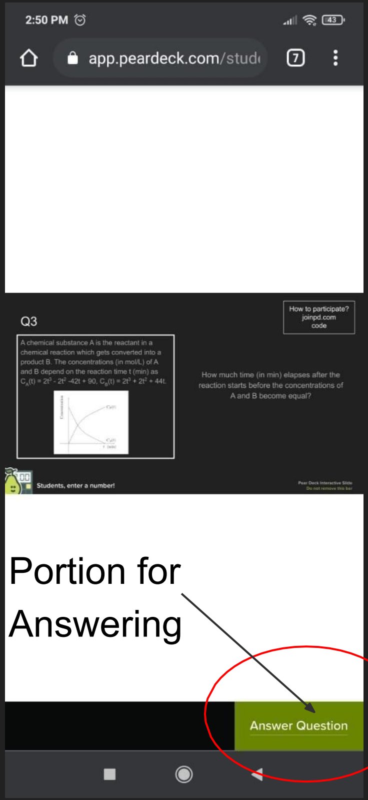
- For every question - 5 to 10 minutes allotted
    - Question will be shown on the screen for solving
    - If you are done solving, then enter your answer on joinpd.com
    - Instructor will provide the solution
    - Questions and discussion

Example Screenshots

Mobile

Laptop/Desktop





Test Problem

How to participate?
joinpd.com
code: see above

This is a problem to test if everything is working.

How many marks do you expect in quiz-1?

Q1

Suppose $2p = q$, where $p,q \in \mathbb{R}$.

Q1

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Solution

- $$\frac{4^p}{2^q} + \frac{25^q}{625^p} = \frac{2^{2p}}{2^q} + \frac{25^q}{25^{2p}}$$
  $$= 2^{2p-q} + 25^{q-2p}$$

Solution

- $\frac{4^p}{2^q} + \frac{25^q}{625^p} = \frac{2^{2p}}{2^q} + \frac{25^q}{25^{2p}}$

  $= 2^{2p-q} + 25^{q-2p}$

- Given, 2p = q ⇒ 2p - q = 0 (or) q - 2p = 0
  substituting these values, we get

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- $2^{2p-q} + 25^{q-2p}$ = 2^0 + 25^0 = 1 + 1 = 2

Solution

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- $2^{2p-q} + 25^{q-2p}$ = 2^0 + 25^0 = 1 + 1 = 2

Answer: 2

Q2

Half-life of an element is the time required for half of a given sample of radioactive element to change to another element. The rate of change of concentration is calculated by the formula $A(t) = A_o(\frac{1}{2})^{(\frac{t}{\gamma})}$ where γ is the half-life of the material, A_o is the initial concentration of the radioactive element in the given sample, $A(t)$ is the concentration of the radioactive element in the sample after time t.

Q2

Half-life of an element is the time required for half of a given sample of radioactive element to change to another element. The rate of change of concentration is calculated by the formula $A(t) = A_o(\frac{1}{2})^{(\frac{t}{\gamma})}$ where γ is the half-life of the material, A_o is the initial concentration of the radioactive element in the given sample, $A(t)$ is the concentration of the radioactive element in the sample after time t.

If Radium has a half-life of 1600 years and the initial concentration of Radium in a sample was 100%, then calculate the percentage of Radium in that sample after 2000 years.

Students, write your response!

Solution

Step 1:

Given $A(t) = A_o(\frac{1}{2})^{(\frac{t}{\gamma})}$ and half life of radium is 1600 i.e $\gamma = 1600$

$$\implies A(2000) = A_o(\frac{1}{2})^{(\frac{2000}{1600})}$$

$$= A_o(\frac{1}{2})^{(\frac{5}{4})}$$

Solution

Step 2:

At initial time$(t = 0)$ the concentration of Radium is 100%

$\implies$ At initial time the concentration of Radium is A_o

Solution

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At initial time($t = 0$) the concentration of Radium is 100%

$\implies$ At initial time the concentration of Radium is A_o

Step-3:

So, the percentage of Radium in that sample after 2000 years is

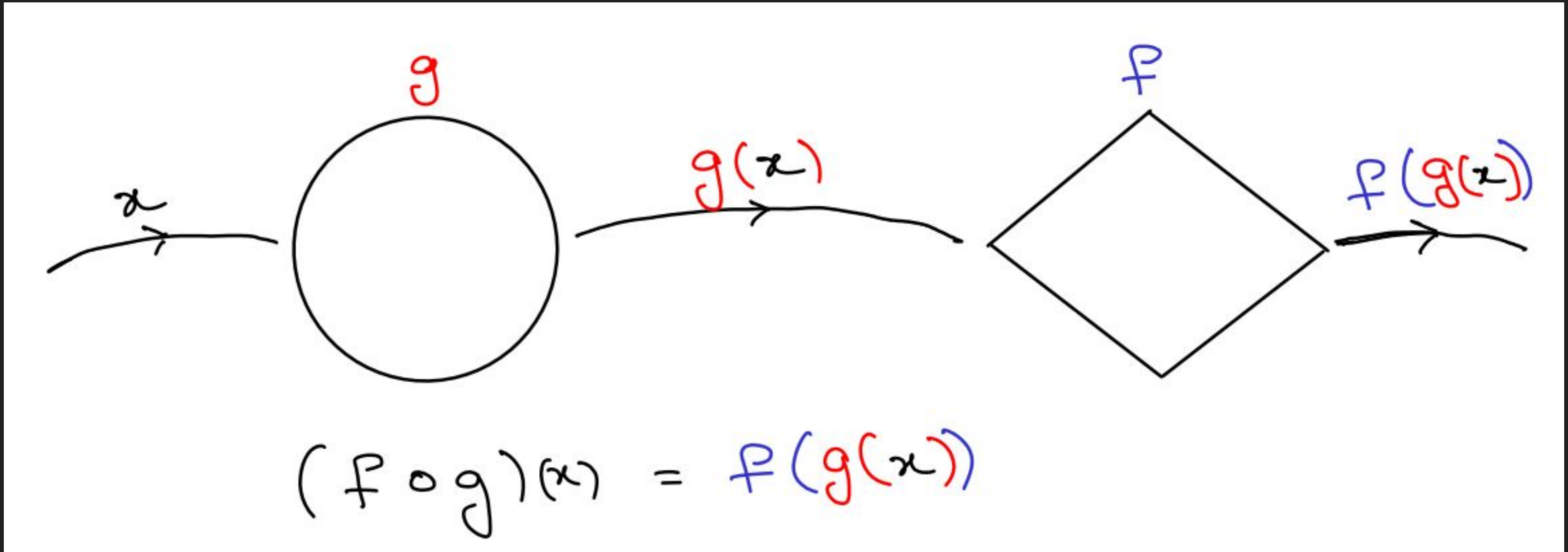
$$\left(\frac{A_o(\frac{1}{2})^{(\frac{5}{4})}}{A_0} \times 100\right)\% = \left(\left(\frac{1}{2}\right)^{(\frac{5}{4})} \times 100\right)\% \approx 42\%$$

Answer: 42

Domain of the composite function (f o g):

The domain of the composite function f o g is the set of all ‘x’ such that

- ‘x’ is in the domain of g.
- g(x) is in the domain of f.



Q3

Two functions $f(x) = (x+1)^{\frac{1}{2}}$ and $g(x) = (x^2 - 1)$ are well defined.

Q3

Two functions $f(x) = (x+1)^{\frac{1}{2}}$ and $g(x) = (x^2 - 1)$ are well defined.

Find the domain of the composite function $g \circ f$.

- ○ $\mathbb{R}$
- ○ $((-3, \infty) \setminus ((-3, 5) \cap (-5, -1)$
- ○ $[-1, \infty)$
- ○ $\mathbb{R} \setminus (1, \infty)$

Students, write your response!

Solution

Step-1: Find the set of all ‘x’ such that ‘x’ is in the domain of the function ‘f’.

Given, $f(x) = (x+1)^{\frac{1}{2}}$.
For $f(x)$ to be well defined, $x + 1 \geq 0 \implies x \geq -1$.
So, the domain of $f(x)$ is $[-1, \infty)$.

Solution

Step-1: Find the set of all ‘x’ such that ‘x’ is in the domain of the function ‘f’.

Given, $f(x) = (x+1)^{\frac{1}{2}}$.
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Step-2: Find the range of f(x) and the domain of the function ‘g’.

Now, the range of $f(x)$ is $[0, \infty)$ and the domain of $g(x) = x^2 - 1$ is $\mathbb{R}$.

Solution

Step-3: As the range of f(x) is subset of the domain of g(x).

Hence, when we use the two conditions to determine the domain of $g \circ f$.
Here, the domain of $g \circ f$ = The domain of $f = [-1, \infty)$.

Solution

Step-3: As the range of f(x) is subset of the domain of g(x).

Hence, when we use the two conditions to determine the domain of $g \circ f$.
Here, the domain of $g \circ f$ = The domain of $f = [-1, \infty)$.

Step-4: Find all the options that represents the interval [-1,∞)

Now, $(-3, \infty) \setminus ((-3, 5) \cap (-5, -1)) = (-3, \infty) \setminus (-3, -1) = [-1, \infty)$.

Answer: (b),(c)

Q4

In a survey, the population growth in an area can be predicted according to the equation $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ where d is the percentage growth rate of population per year and T is the time since the initial population count α_o was taken.

Q4

In a survey, the population growth in an area can be predicted according to the equation $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ where d is the percentage growth rate of population per year and T is the time since the initial population count α_o was taken.

If in 2015, the population of Adyar was 30,000 and the population growth rate is 4% per year, then what will be the approximate population of Adyar in 2020?($T = 0$ corresponds to the year 2015, $T = 1$ corresponds to the year 2016 and so on..)

Solution

Step-1:

Given $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ and initial population count $\alpha_o = 30000$.
The population growth rate $d = 4\%$ per year and $T = 5$.

Solution

Step-1:

Given $\alpha(T) = \alpha_o(1 + \frac{d}{100})^T$ and initial population count $\alpha_o = 30000$. The population growth rate $d = 4\%$ per year and $T = 5$.

Step-2:

Hence, the approximate population of Adyar in 2020 is $\alpha(5) = 30000 \times (1 + \frac{4}{100})^5 \approx 36500$

Answer: 36500

Q5

If $f(x) = x^2$ in the restricted domain $[0,\infty)$, then choose the points where the graphs of the functions $f(x)$ and $f^{-1}(x)$ intersect each other?

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- ○ (0,0)
- ○ (1,1)
- ○ (-1,-1)
- ○ (2,4)

Solution:

Step-1: Find the inverse function $f^{-1}(x)$.

Let us consider a function $g(x) = x^{\frac{1}{2}}$.
Now, $f \circ g = (x^{\frac{1}{2}})^2 = |x| = x$ and $g \circ f = (x^2)^{\frac{1}{2}} = |x| = x$.
($|x| = x$ because restricted domain is $[0,\infty)$).

Hence, g is the inverse function of f.
So, $f^{-1}(x) = x^{\frac{1}{2}} = \sqrt{x}$.

Solution:

Step-2: Find the x-coordinates of the intersection points by equating f and f^{-1}.

To get intersection points

$$\begin{aligned}
& f = f^{-1} \\
\implies & x^2 = x^{\frac{1}{2}} \\
\implies & x^4 = x \\
\implies & x^4 - x = 0 \\
\implies & x(x^3 - 1) = 0 \\
\implies & x((x-1)(x^2+x+1)) = 0 \\
\implies & x(x-1)\underbrace{(x^2+x+1)}_{\text{No real roots}} = 0
\end{aligned}$$

$\therefore\ x = 0, 1$

Solution:

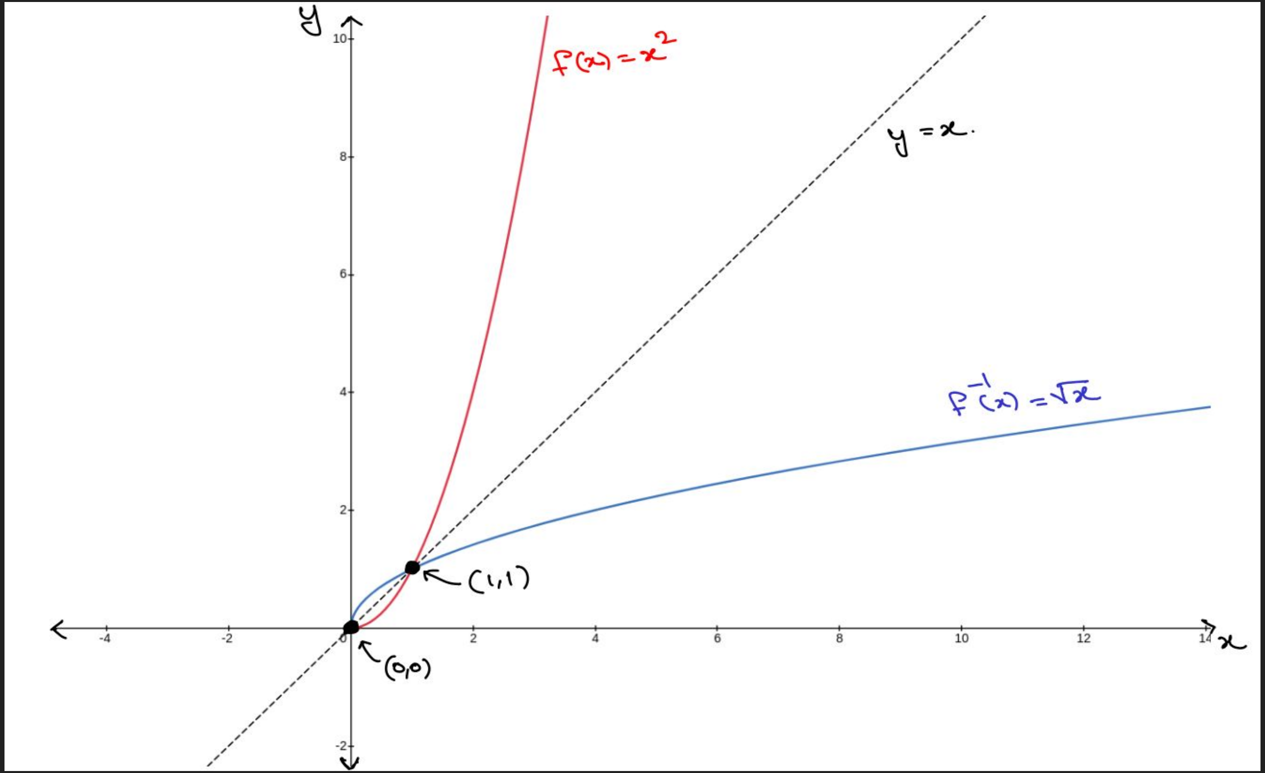
Step-3: Find the intersection points.

- The points are (0,f(0)) and (1,f(1)).
- f(0) = 0 and f(1) = 1
- Hence, the intersection points are (0,0) and (1,1).

Answer: (a), (b)

Solution:

Alternate step: From the graphs of f and f^{-1}.



THANK YOU!!